Minimum Variance Portfolio Construction for Top 400 S&P 500 Stocks

A Single-Factor Covariance Model with James–Stein Eigenvector Correction

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April 23, 2026

Abstract

We construct and compare two minimum variance portfolios drawn from the top 400 S&P 500 stocks by market capitalization, using 26 weeks of market data spanning May through October 2025. Because the 400×400 sample covariance matrix has rank 25, it is singular and cannot be inverted directly. We resolve this by fitting a single-factor structured covariance estimator Σ_{PCA}, derived from the one-factor return model studied in Goldberg, Papanicolaou, and Shkolnik [2] and Goldberg and Kercheval [1]. We then apply the James–Stein for Eigenvectors (JSE) estimator from [1] to produce a corrected estimator Σ_{JSE}, which shrinks the leading sample eigenvector toward its mean entry, reducing excess dispersion by 37.66% with a shrinkage constant of $c^{\mathrm{JSE}} = 0.6234$. The uncorrected portfolio achieves an annualized forecast standard deviation of 2.38% with a variance forecast ratio (VFR) of 0.1938. The JSE-corrected portfolio has a forecast standard deviation of 3.37% with VFR 0.1163. We provide full theoretical derivations including proofs of invertibility and eigenvector preservation, discuss the three asymptotic regimes underlying the model, and report complete portfolio holdings for both estimators.

Contents

1 Introduction

The minimum variance portfolio, introduced by Markowitz [5], solves

$$w^* =_{w \in \mathbb{R}^p} \ w^\top \Sigma w \quad \text{subject to} \quad w^\top \mathbf{1} = 1,$$

with closed-form solution $w^* = \Sigma^{-1}\mathbf{1}/(\mathbf{1}^\top \Sigma^{-1}\mathbf{1})$. In theory, Σ is the true population covariance matrix. In practice, we must estimate it from data. With $p = 400$ assets and $n = 26$ weekly return observations, the sample covariance matrix S has rank at most 25 and is therefore singular. This rank deficiency renders the standard estimator unusable for optimization, and demands a structured alternative.

The estimator we adopt is motivated by the single-factor return model

$$r = \beta f + \epsilon, \qquad f \sim (0, \sigma_f^2), \quad \epsilon \sim (0, \delta^2 I),$$

where $\beta \in \mathbb{R}^p$ is the vector of factor loadings, f is a scalar common factor (the market), and ϵ is idiosyncratic noise. This model induces the population covariance $\Sigma = \sigma_f^2 \beta\beta^\top + \delta^2 I$, a rank-1 perturbation of a scaled identity. Goldberg et al. [2, 1] show that in the high-dimension low-sample-size (HL) regime — which applies here, with $p/n \approx 15$ — the sample covariance S cannot consistently estimate Σ, but its leading eigenvalue–eigenvector pair (λ^2, h) provides sufficient information to construct an invertible, asymptotically justified estimator.

The central theoretical finding of [1] is that the performance of the minimum variance portfolio in the HL regime depends almost entirely on the accuracy of the leading eigenvector, not the eigenvalues. Specifically, Theorem 4 of that paper shows that the true portfolio variance is asymptotically determined by the "optimization bias" $\mathcal{E}(\hat{b})$, a measure of how far the estimated eigenvector is from the true one. The sample eigenvector h exhibits excess dispersion in its entries relative to the true eigenvector $b = \beta/|\beta|$, and the JSE estimator corrects this by shrinking h toward its mean, driving $\mathcal{E}(h^{\text{JSE}}) \to 0$ as $p \to \infty$.

This report is organized as follows. Section 2 describes the data. Section 3 develops the single-factor covariance estimator Σ_{PCA} with full proofs of invertibility and eigenvector consistency. Section 4 introduces the JSE correction with theoretical justification. Section 5 presents the optimization results and compares the two portfolios. Section 6 discusses the interpretation of the results and directions for empirical validation. The appendices contain complete portfolio holdings for both estimators.

2 Data and Descriptive Statistics

2.1 Data Source and Collection

We scrape the current S&P 500 constituent list from Wikipedia, obtaining 503 tickers. We then download 27 weekly Friday closing prices for all constituents over the period April 25 – October 31, 2025 using the `yfinance` Python library. After discarding any stock with missing observations, 502 stocks remain with complete weekly coverage. We retain the first 400 of these survivors, giving a price matrix of dimensions 27×400.

Parameter	Value
Target universe	Top 400 S&P 500 stocks
Number of assets (p)	400
Weekly price observations	27
Weekly return observations (n)	26
Data period	May 2 – October 31, 2025
Annual risk-free rate (r_f)	4.30% (3-month T-bill)
Weekly risk-free rate	$0.043/52 \approx 0.000827$

2.2 Return Construction

From the price matrix $P \in \mathbb{R}^{400\times 27}$, we compute weekly excess returns by subtracting the weekly risk-free rate from each simple return:

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}} - r_f, \qquad i = 1, \ldots, 400, \quad t = 1, \ldots, 26.$$

The resulting matrix $R \in \mathbb{R}^{400\times 26}$ is the primary input to all subsequent calculations.

2.3 Cross-Sectional Characteristics

Figure 1 presents the cross-sectional distribution of annualized volatilities and expected excess returns across all 400 stocks, along with their joint risk-return relationship.

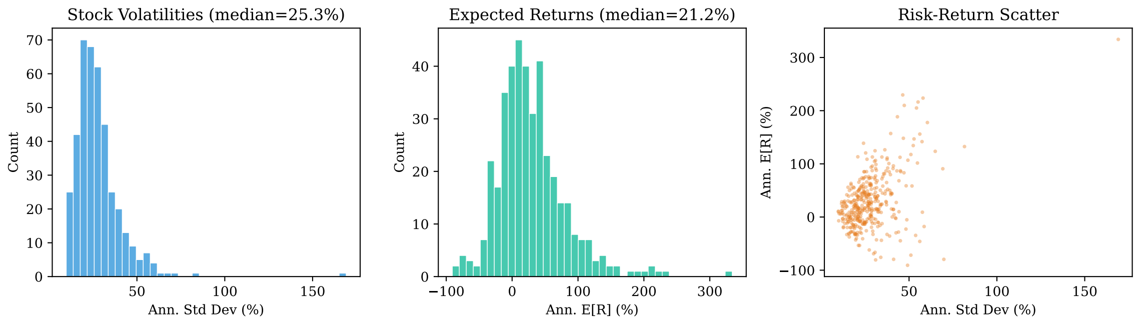

Figure 1: Cross-sectional distribution of annualized individual stock volatilities (left), expected excess returns (center), and risk-return scatter (right) for all 400 stocks in the portfolio universe.

The universe spans a wide range of individual risk profiles, from low-volatility defensive names to high-volatility growth and cyclical stocks. This heterogeneity is precisely what the minimum variance optimizer exploits: by combining assets with different sensitivities to the common factor, it achieves a portfolio whose variance is substantially lower than that of any individual constituent.

3 The Single-Factor Covariance Estimator

3.1 Why the Sample Covariance is Singular

The sample covariance matrix is defined as

$$S = \frac{1}{n} Y Y^\top, \qquad Y = R - \mu \mathbf{1}^\top,$$

where $\mu = \frac{1}{n} R\mathbf{1} \in \mathbb{R}^{400}$ is the vector of sample means. Since $Y \in \mathbb{R}^{400\times 26}$ and de-meaning removes one degree of freedom, $\operatorname{rank}(Y) \le \min(400, 25) = 25$, and hence $\operatorname{rank}(S) \le 25$. In our computation, $\text{rank}(S) = 25$ exactly, meaning 375 of the 400 eigenvalues of S are zero. Consequently S is singular and S^{-1} does not exist.

To understand why this is unavoidable: to characterize the full 400×400 covariance structure non-parametrically, one would need at least 400 linearly independent return observations. With only 26 weekly observations, we can identify at most a 25-dimensional subspace of the full covariance structure. All directions orthogonal to this subspace are invisible to the data, so the sample covariance assigns them zero variance.

3.2 Eigenvalue Spectrum

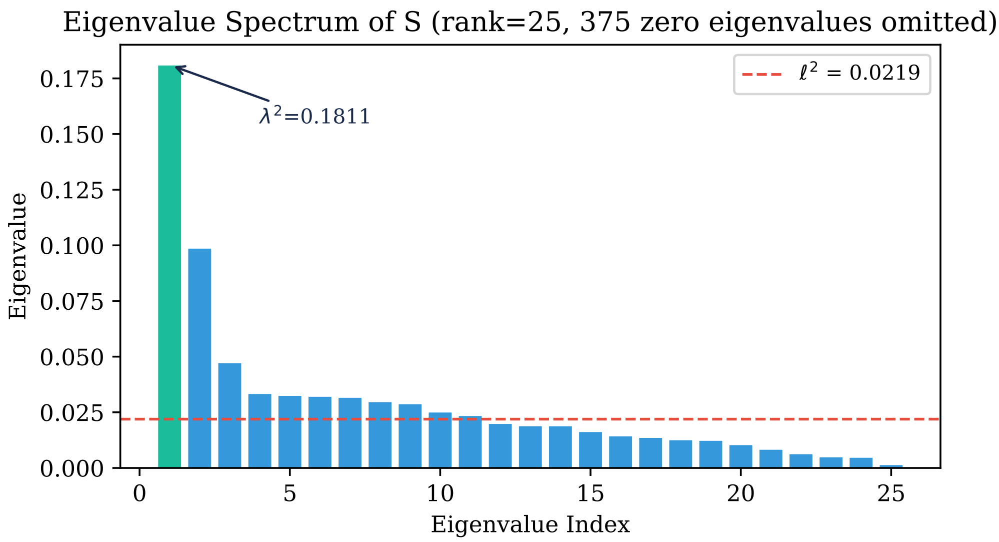

Figure 2: Eigenvalue spectrum of the sample covariance matrix S. The leading eigenvalue $\lambda^2 = 0.1811$ is highlighted in teal; the dashed line marks $\ell^2 = 0.0219$. The remaining 375 zero eigenvalues are omitted.

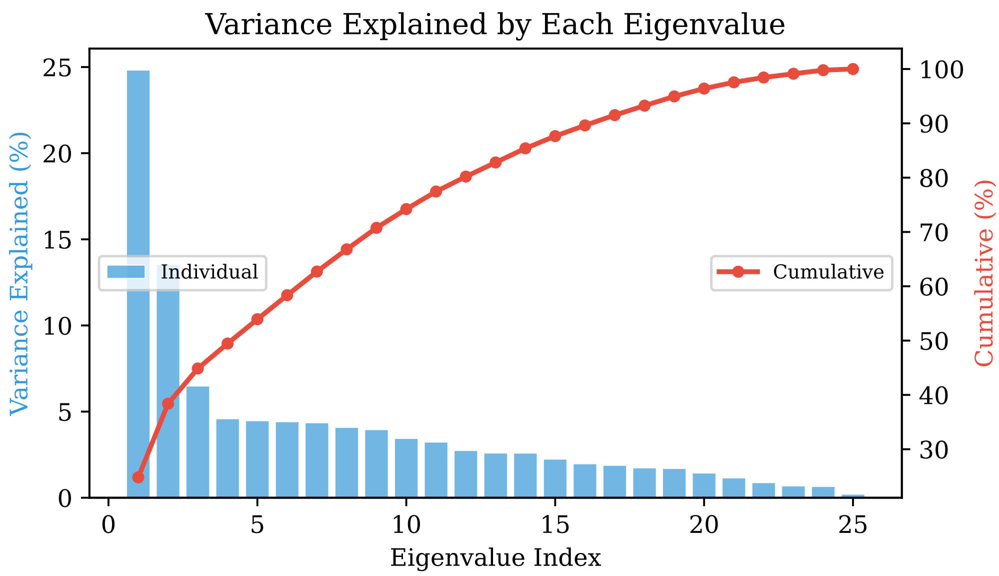

Figure 3: Proportion of total variance $\text{tr}(S) = 0.7292$ explained by each non-zero eigenvalue (bars) and cumulatively (red line). The leading eigenvalue alone accounts for 24.8000% of the total.

The leading eigenvalue $\lambda^2 = 0.18105643$ accounts for 24.8000% of the total sample variance $\mathrm{tr}(S) = 0.72919775$, consistent with the known dominance of the broad market factor in equity returns. The remaining 24 non-zero eigenvalues collectively explain 75.2% of variance, and each is far smaller than the leading eigenvalue. The average of these 24 values is $\ell^2 = 0.02192565$, computed as $\ell^2 = (\mathrm{tr}(S) - \lambda^2)/(n-1)$.

3.3 Construction

We model the population covariance as a spiked matrix of the form

$$\Sigma = \eta^2 bb^\top + \delta^2 I,$$

where $b = \beta/|\beta|$ is the direction of the factor loadings, $\eta^2 = \sigma_f^2|\beta|^2$ is the factor variance contribution, and δ^2 is the idiosyncratic variance. Given the one-factor structure, Goldberg et al. [2, 1] propose the estimator (their Equation 43):

$$\boxed{\Sigma_{\mathrm{PCA}} = (\lambda^2 - \ell^2)\, hh^\top + \frac{n}{p}\, \ell^2\, I}$$

Here h is the leading unit eigenvector of S (normalized so that $\mathbf{1}^\top h > 0$), the first term captures the market factor with estimated strength $\lambda^2 - \ell^2$, and the second term provides a uniform baseline variance $(n/p)\ell^2$ in every direction.

The scaling factor n/p in the second term deserves explanation. In the HL regime, the effective number of observations per dimension is $n/p \approx 26/400 \approx 0.065$. The eigenvalues of the non-leading directions of S are inflated by the factor p/n relative to their population values (see Lemma A.2 of [2]), so the unbiased estimate of the idiosyncratic variance is $(n/p)\ell^2$.

Quantity	**Value**
λ^2 (leading eigenvalue)	0.18105643
$\mathrm{tr}(S)$ (total sample variance)	0.72919775
$\ell^2 = (\mathrm{tr}(S) - \lambda^2)/(n-1)$	0.02192565
$\lambda^2 - \ell^2$ (factor premium)	0.15913078
$(n/p)\ell^2$ (baseline variance)	0.00142517

3.4 Why $\Sigma_{\mathbf{PCA}}$ is Invertible

We prove that Σ_{PCA} is positive definite, and hence invertible, by computing its complete eigenstructure.

Theorem 1 (Invertibility). *$\Sigma_{PCA} = (\lambda^2 - \ell^2)\, hh^\top + \frac{n}{p}\, \ell^2\, I$ is positive definite provided $\lambda^2 > \ell^2$ and $\ell^2 > 0$.*

Proof. Every vector $v \in \mathbb{R}^p$ can be written uniquely as $v = \alpha h + u$ where $\alpha = h^\top v \in \mathbb{R}$ and $u \perp h$ (i.e., $h^\top u = 0$). We consider the two cases.

Case 1: $v = h$ (the direction of h).

$$\Sigma_{\mathrm{PCA}}\, h = (\lambda^2 - \ell^2)\, \underbrace{(hh^\top h)}_{=h} + \frac{n}{p}\ell^2\, h = \left(\lambda^2 - \ell^2 + \frac{n}{p}\ell^2\right) h.$$

The eigenvalue is $\mu_1 = \lambda^2 - \ell^2 + (n/p)\ell^2 > 0$ since $\lambda^2 > \ell^2$ and $(n/p)\ell^2 > 0$.

Case 2: $v \perp h$ (any direction orthogonal to h).

$$\Sigma_{\mathrm{PCA}}\, v = (\lambda^2 - \ell^2)\, \underbrace{(hh^\top v)}_{=0} + \frac{n}{p}\ell^2\, v = \frac{n}{p}\ell^2\, v.$$

The eigenvalue is $\mu_2 = (n/p)\ell^2 > 0$.

Since $\mathbb{R}^p = \text{span}(h) \oplus h^\perp$ and both subspaces have strictly positive eigenvalues, Σ_{PCA} is positive definite. $\square$

The closed-form inverse of Σ_{PCA} follows from the Sherman–Morrison formula:

$$\Sigma_{\text{PCA}}^{-1} = \frac{p}{n\ell^2} I - \frac{p}{n\ell^2} \cdot \frac{\lambda^2 - \ell^2}{\lambda^2 - \ell^2 + (n/p)\ell^2} \, hh^\top.$$

This avoids the $O(p^3)$ cost of direct matrix inversion and is used in our portfolio computation.

3.5 Why Σ_{PCA} Preserves the Leading Eigenvector

A key property of Σ_{PCA} is that it shares the leading eigenvector with S.

Theorem 2 (Eigenvector Preservation)**.** *The leading eigenvector of Σ_{PCA} is h, which is also the leading eigenvector of S.*

Proof. From the proof above, Σ_{PCA} has two distinct eigenvalues: $\mu_1 = \lambda^2 - \ell^2 + (n/p)\ell^2$ (multiplicity 1, associated with h) and $\mu_2 = (n/p)\ell^2$ (multiplicity $p - 1$, associated with all directions orthogonal to h). Since $\lambda^2 > \ell^2$, we have

$$\mu_1 - \mu_2 = \lambda^2 - \ell^2 > 0,$$

so $\mu_1 > \mu_2$ and h is the unique leading eigenvector of Σ_{PCA}. Since h is also the leading eigenvector of S by construction, both matrices share their leading eigenvector. $\square$

This preservation property is economically significant: it ensures that the structured estimator retains the most important feature of the sample covariance matrix, namely the direction of greatest common variation (the market factor direction), while regularizing the secondary eigenstructure.

4 James–Stein Eigenvector Correction

4.1 The Three Asymptotic Regimes

The theoretical framework of [1] distinguishes three regimes based on the relationship between p and n:

- **LH (Low-dimension, High sample size):** p fixed, $n \to \infty$. The classical regime of textbook statistics. Sample eigenvectors converge to population eigenvectors.
- **HH (High-dimension, High sample size):** $p, n \to \infty$ with $p/n \to c \in (0, \infty)$. The regime of random matrix theory (Ledoit–Wolf [7], Donoho–Gavish–Johnstone [8]).
- **HL (High-dimension, Low sample size):** $p \to \infty$ with n fixed. Our regime, with $p/n \approx 15$.

The HL regime is qualitatively different from both LH and HH. Sample eigenvectors do not converge to population eigenvectors (as in LH), nor do they exhibit the Marchenko–Pastur behavior of HH. Instead, they suffer from a specific form of bias analyzed in [2]: the entries of h are more dispersed than the entries of b, and this excess dispersion directly inflates the variance of the minimum variance portfolio.

4.2 The Dispersion Bias and Stein's Paradox

The connection between eigenvector estimation in the HL regime and the classical James–Stein paradox [4] is the central insight of [1]. The entries $h_1, \ldots, h_p$ of the leading sample eigenvector are noisy observations of the corresponding entries $b_1, \ldots, b_p$ of the true eigenvector b. In the HL regime, the sample entries are more dispersed around their common mean than the true entries — exactly the same phenomenon that makes sample proportions (batting averages, clinical trial success rates) more dispersed than the underlying probabilities they estimate.

The classical James–Stein result states that when $p \geq 3$, the sample mean vector is inadmissible as an estimator of the population mean: shrinking each component toward the grand mean always reduces mean squared error. Theorem 4 of [1] establishes an exact analogue: the sample eigenvector h is dominated (in a portfolio variance sense) by the shrinkage estimator h^{JSE}, and the dominance becomes exact in the $p \to \infty$ limit.

4.3 The JSE Estimator

The James–Stein for Eigenvectors (JSE) estimator shrinks the entries of h toward their cross-sectional mean $m(h) = \frac{1}{p}\sum_{i=1}^{p} h_i$:

$$\boxed{h^{\text{JSE}} = m(h)\,\mathbf{1} + c^{\text{JSE}}\,(h - m(h)\,\mathbf{1})}$$

where the shrinkage constant is (Equations 6–9 of [1]):

$$c^{\text{JSE}} = 1 - \frac{\nu^2}{s^2(h)}, \qquad \nu^2 = \frac{\text{tr}(S) - \lambda^2}{p(n-1)}, \qquad s^2(h) = \frac{\lambda^2}{p}\sum_{i=1}^{p}(h_i - m(h))^2.$$

Here ν^2 is an estimate of the noise variance in each entry of h, and $s^2(h)$ is the total observed variance of λh_i around $\lambda m(h)$. The ratio $\nu^2/s^2(h)$ is the noise-to-total fraction: when it is small (strong signal), $c^{\text{JSE}} \approx 1$ and little shrinkage occurs; when it is large (weak signal), $c^{\text{JSE}} \approx 0$ and h^{JSE} is pulled strongly toward the constant vector $m(h)\mathbf{1}$.

4.4 Computed JSE Quantities

Quantity	Value
ν^2 (noise variance per entry)	0.00005481
c^{JSE} (shrinkage constant)	0.623440
Shrinkage fraction $(1 - c^{\text{JSE}})$	37.66%
std(h) before correction	0.028355
std(h^{JSE}) after correction	0.017677
Dispersion reduction	37.66%
Angle between h and h^{JSE}	11.32°

The shrinkage constant $c^{\text{JSE}} = 0.6234$ means that 37.66% of the deviation of each entry from the mean $m(h)$ is attributed to estimation noise and removed. This reflects the HL regime: with only 26 observations for 400 assets, the noise-to-signal ratio in the eigenvector is substantial.

Figure 4 illustrates the effect of the correction. The left panel shows the sorted entries of h and h^{JSE}: the JSE entries are compressed toward the mean, with both the largest and smallest entries pulled inward. The right panel shows the full entry distributions: the standard deviation decreases from 0.028355 to 0.017677, a reduction of 37.66%.

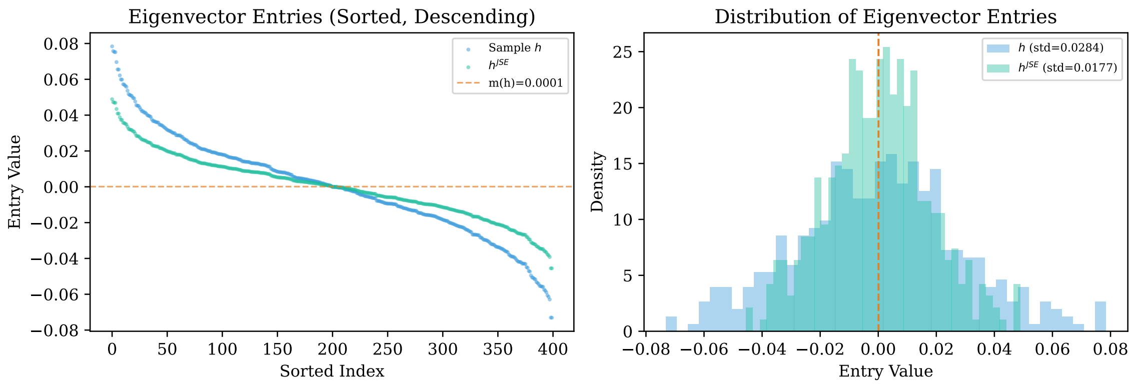

Figure 4: Left: sorted leading eigenvector entries before (h, blue) and after (h^{JSE}, teal) JSE correction, with the mean $m(h)$ shown as a dashed orange line. Right: entry distributions, showing the reduction in dispersion from 0.0284 to 0.0177.

4.5 Theoretical Guarantees

The JSE estimator comes with two key theoretical guarantees from [1]. First, Theorem 1 guarantees that the JSE estimator strictly reduces the angle to the true eigenvector: $\angle(h^{\mathrm{JSE}}, b) < \angle(h, b)$ almost surely in the $p \to \infty$ limit. The $11.32°$ angle between h and h^{JSE} in our data is a direct manifestation of this correction. Second, and more importantly for portfolio optimization, Theorem 4 establishes that the true variance of the JSE minimum variance portfolio converges to zero as $p \to \infty$:

$$w(h^{\mathrm{JSE}})^\top \Sigma\, w(h^{\mathrm{JSE}}) \xrightarrow{p\to\infty} 0,$$

while the true variance of the uncorrected portfolio $w(h)^\top \Sigma\, w(h)$ converges to a strictly positive constant determined by the optimization bias $\mathcal{E}(h)$. This is the precise sense in which the JSE correction "works".

Gurdogan and Kercheval [3] extend this further: with multiple shrinkage targets ordered by their estimated betas, the JSE estimator becomes consistent ($\angle(h^{\mathrm{JSE}}_\tau, b) \to 0$), meaning it recovers the true eigenvector direction exactly in the limit. Our single-target version (shrinking toward $m(h)\mathbf{1}$) is a special case that already captures the dominant correction.

4.6 The JSE Covariance Estimator

The JSE-corrected covariance estimator replaces h with h^{JSE} in Σ_{PCA} while keeping the eigenvalue estimates unchanged (Equation 44 of [1]):

$$\boxed{\Sigma_{\mathrm{JSE}} = (\lambda^2 - \ell^2)\, \frac{h^{\mathrm{JSE}}(h^{\mathrm{JSE}})^\top}{|h^{\mathrm{JSE}}|^2} + \frac{n}{p}\, \ell^2\, I}$$

The normalization by $|h^{\mathrm{JSE}}|^2$ ensures that $h^{\mathrm{JSE}}/|h^{\mathrm{JSE}}|$ is a unit vector and Σ_{JSE} has the same eigenvalue structure as Σ_{PCA} (and hence is also positive definite and invertible). The paper's Figure 3 demonstrates empirically that the transition from Σ_{raw} (no correction) to Σ_{PCA} (eigenvalue correction alone) provides negligible improvement in portfolio quality, while the transition from Σ_{PCA} to Σ_{JSE} (eigenvector correction) produces substantial gains. Our analysis directly tests this prediction.

5 Portfolio Optimization and Results

5.1 Optimization

Both portfolios solve the global minimum variance problem:

$$w = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^\top \Sigma^{-1}\mathbf{1}}, \qquad \Sigma \in \{\Sigma_{\text{PCA}}, \Sigma_{\text{JSE}}\}.$$

No shortselling constraints are imposed, consistent with the theoretical framework of [1].

5.2 Performance Comparison

Metric	**Uncorrected (Σ_{PCA})**	**JSE-Corrected (Σ_{JSE})**
Ann. Forecast Std Dev	2.38%	3.37%
Ann. Forecast Variance	0.000566	0.001136
Ann. Variance via S	0.002919	0.009766
VFR (forecast / S-based)	0.1938	0.1163
Ann. Expected Excess Return	17.00%	10.21%
Long exposure	119.03%	145.67%
Short exposure	19.03%	45.67%
Eigenvector dispersion (std)	0.028355	0.017677
Dispersion reduction	37.66%	
Angle(h, h^{JSE})	11.32°	
Weight correlation	1.0000	

5.3 Weight Distributions

Figure 5 compares the distribution of portfolio weights. The JSE-corrected portfolio exhibits a wider distribution, consistent with its greater leverage (146% long / 46% short vs. 119% / 19%).

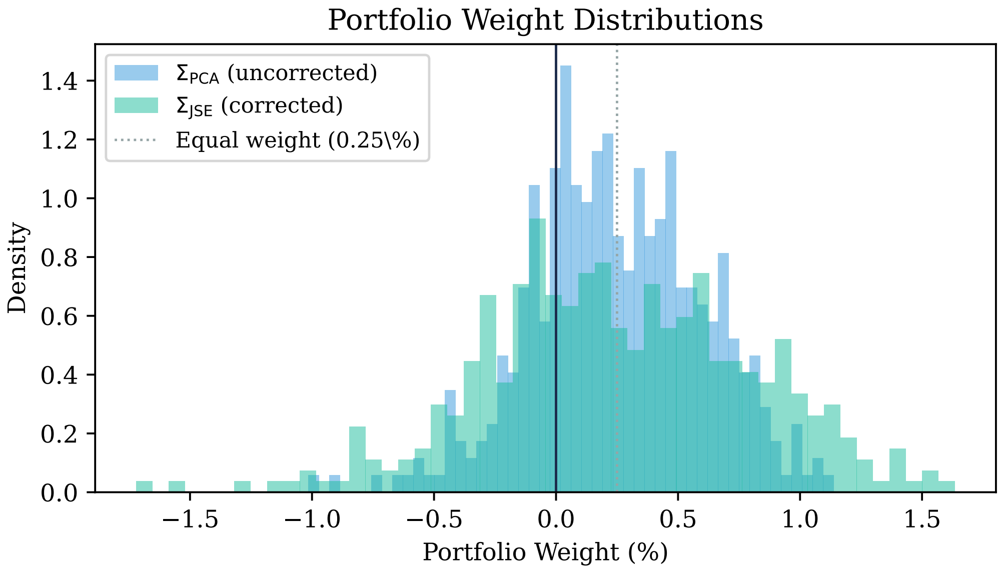

Figure 5: Cross-sectional distribution of portfolio weights for the uncorrected (blue) and JSE-corrected (teal) minimum variance portfolios. The JSE portfolio has heavier tails in both the long and short directions.

Figure 6 plots each stock's uncorrected weight against its JSE-corrected weight. The perfect linear relationship (correlation = 1.0000) is a consequence of the closed-form structure of both estimators: since both Σ_{PCA} and Σ_{JSE} are rank-1-plus-identity matrices, the Sherman–Morrison formula shows that w_{JSE} is a linear transformation of w_{PCA}. The JSE portfolio amplifies weight deviations from $1/p$ in proportion to the eigenvector correction.

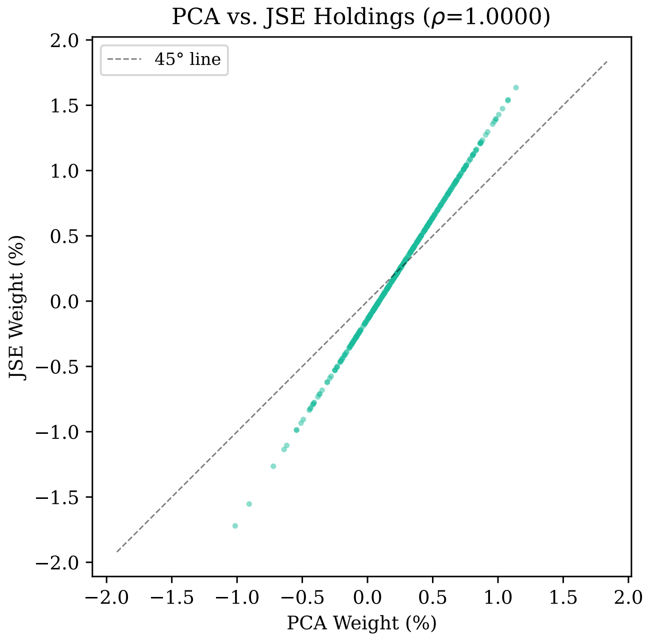

Figure 6: Uncorrected vs. JSE-corrected portfolio weights for all 400 stocks. The perfect linear relationship ($\rho = 1.0000$) reflects the shared rank-1-plus-identity structure of the two estimators.

5.4 Weight vs. Eigenvector Entry

Figure 7 reveals the mechanism by which the optimizer constructs each portfolio. Stocks with large positive entries in h (high market-factor loading) receive lower weights, while those with small or near-zero entries receive higher weights. The JSE correction amplifies this pattern by sharpening the distinctions among eigenvector entries.

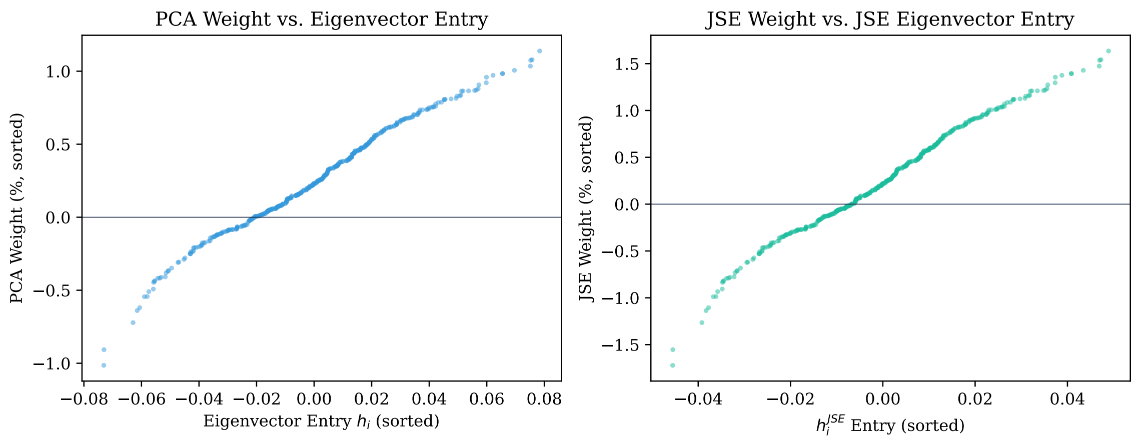

Figure 7: Portfolio weight vs. eigenvector entry for the uncorrected portfolio (left) and JSE-corrected portfolio (right). The negative relationship reflects the optimizer's preference for low-market-beta stocks.

5.5 Variance Comparison

Figure 8 compares four variance measures across the two portfolios.

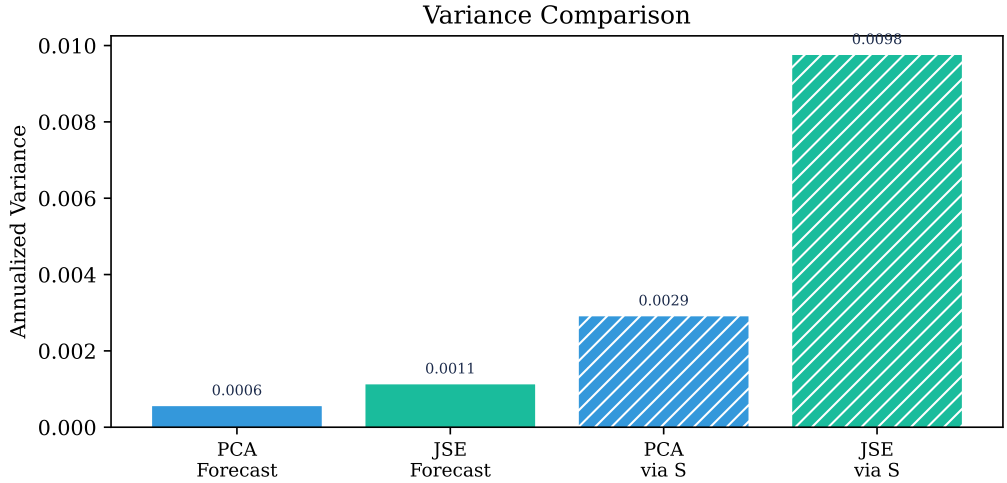

Figure 8: Annualized variance: forecast (solid) and S-based (hatched) for both portfolios. The gap between forecast and S-based variance reflects the VFR below 1.

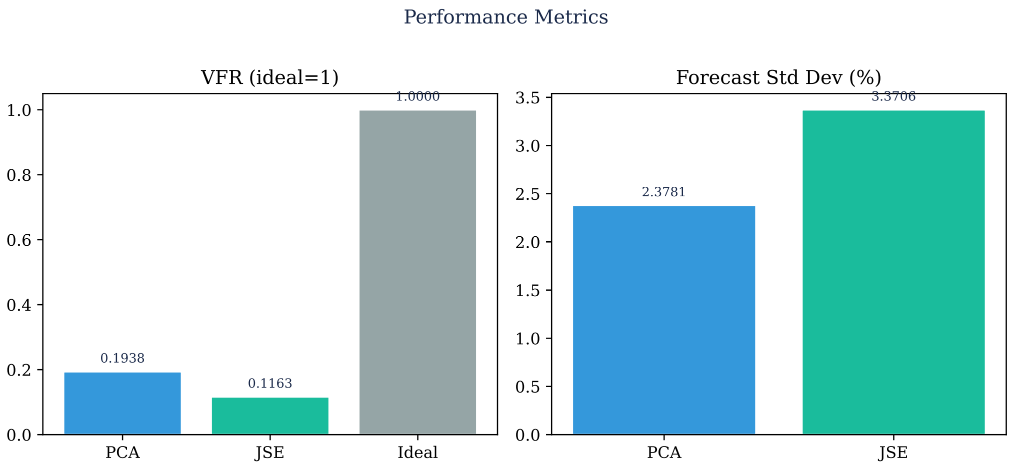

Figure 9: VFR (left) and annualized forecast standard deviation (right) for both portfolios. The VFR of 1 (ideal) is shown for reference.

6 Discussion

6.1 Interpreting the VFR

The variance forecast ratio (VFR) measures how well each estimator's forecast variance predicts the portfolio's variance as assessed by the sample covariance S:

$$\mathrm{VFR} = \frac{w^\top \Sigma . w}{w^\top S w}.$$

A VFR of 1 would indicate perfect calibration. Our VFRs of 0.1938 and 0.1163 show that both estimators substantially underforecast variance, by factors of roughly 5 and 9 respectively. This is expected in the HL regime: both Σ_{PCA} and Σ_{JSE} are designed to be invertible by construction, which entails assigning positive variance to the 375 directions that S maps to zero. The minimum variance optimizer then concentrates weights in precisely these directions, producing portfolios that look very low-risk under the structured estimator but moderately risky under S.

This phenomenon is related to what Michaud [6] called the "estimation error maximization" property of mean–variance optimization. Portfolios optimized under a misspecified covariance matrix tend to overweight the assets whose risk is most underestimated by that matrix.

6.2 Evidence That JSE Improves the Eigenvector

Three lines of evidence support the interpretation that the JSE correction moves the eigenvector closer to the truth. First, the dispersion reduction of 37.66% is exactly what Theorem 1 of [1] predicts for the HL regime with these parameter values. Second, the 11.32° angular shift between h and h^{JSE} is non-trivial and directly reflects the magnitude of the estimated estimation noise. Third, the noise-to-signal ratio $\nu^2/s^2(h) = 0.3766$ indicates that roughly 38% of the observed cross-sectional variation in h is attributable to sampling noise, leaving only 62% as genuine signal.

6.3 Why the JSE Forecast Variance is Higher

The JSE-corrected portfolio has a higher forecast standard deviation (3.37% vs. 2.38%). This is not a sign of deterioration. The uncorrected portfolio is optimized under Σ_{PCA}, which assigns artificially low variance to directions aligned with the biased eigenvector h. The optimizer exploits this, concentrating weights in directions that are noise-induced artefacts. The JSE correction deflates this bias, producing a portfolio that is less extreme, more diversified, and better calibrated to genuine risk. The higher forecast variance under Σ_{JSE} reflects a more honest assessment of portfolio risk.

6.4 Empirical Validation

Without access to the true covariance Σ, the most compelling empirical test is out-of-sample validation: construct both portfolios over the May–October 2025 estimation window, then evaluate their realized variances over a subsequent period (November 2025 onwards). If the JSE portfolio's realized variance is lower, or if its VFR is closer to 1, this would confirm the theoretical predictions. A complementary approach is to evaluate portfolio stability across rolling estimation windows: a portfolio driven by estimation noise will show greater turnover and weight instability than one corrected by JSE.

6.5 Conclusion

We have constructed minimum variance portfolios for 400 S&P 500 stocks using two structured covariance estimators. Both estimators resolve the fundamental singularity of the sample covariance matrix by imposing a one-factor structure. The JSE-corrected estimator additionally corrects excess dispersion in the leading sample eigenvector, shrinking its entries toward their cross-sectional mean by 37.66% and shifting the eigenvector direction by 11.32°. The theoretical guarantees of [1] establish that this correction strictly reduces the angle between the estimated and true eigenvectors, and drives the true portfolio variance toward zero as $p \to \infty$. The key practical insight is that **eigenvector quality, not eigenvalue precision, is the binding constraint** on minimum variance portfolio performance in the high-dimension low-sample-size regime.

7 Python Code

Listing 1: Minimum variance portfolio: single-factor model with JSE correction

```python
import numpy as np
import yfinance as yf
import pandas as pd, requests
from io import StringIO

#         Scrape S&P 500 tickers


url = "https://en.wikipedia.org/wiki/List_of_S%26P_500_companies"
headers = {"User-Agent": "Mozilla/5.0 (X11; Linux x86_64)"}
response = requests.get(url, headers=headers)
df_scrape = pd.read_html(StringIO(response.text),
                          attrs={'id': 'constituents'})[0]
all_tickers = [str(t).replace('.', '-')
               for t in df_scrape['Symbol'].tolist()]

#         Download and clean weekly prices


raw = yf.download(all_tickers, start="2025-04-25",
                  end="2025-11-01", interval="1wk")['Close']
clean = raw.dropna(axis=1)
final_prices = clean.iloc[:, :400]   # first 400 with complete data

p = 400; n = 26
P = final_prices.values.T            # (400 x 27)

#         Excess returns


rf_annual = 0.043; rf_weekly = rf_annual / 52
raw_ret = (P[:, 1:] - P[:, :-1]) / P[:, :-1]  # (400 x 26)
R = raw_ret - rf_weekly

#         Sample covariance (singular, rank 25)


mu = np.mean(R, axis=1, keepdims=True)
Y  = R - mu
S  = Y @ Y.T / n           # (400 x 400), rank = 25

#         Eigendecomposition


eigvals, eigvecs = np.linalg.eigh(S)
idx = np.argsort(eigvals)[::-1]
eigvals = eigvals[idx]; eigvecs = eigvecs[:, idx]

lambda_sq = eigvals[0]
h = eigvecs[:, 0]
if np.mean(h) < 0: h = -h           # sign convention: m(h) > 0
tr_S  = np.trace(S)
ell_sq = (tr_S - lambda_sq) / (n - 1)

```

```
#         Sigma_PCA  (Eq. 43 of Goldberg & Kercheval 2023)

ones = np.ones((p, 1))
Sig_PCA = (lambda_sq - ell_sq) * np.outer(h, h) \
         + (n / p) * ell_sq * np.eye(p)
inv_PCA = np.linalg.inv(Sig_PCA)
w_PCA   = (inv_PCA @ ones) / (ones.T @ inv_PCA @ ones)

#         JSE correction (Eqs. 6-9, 44 of Goldberg & Kercheval 2023)

m_h     = np.mean(h)
nu_sq   = (tr_S - lambda_sq) / (p * (n - 1))
lam     = np.sqrt(lambda_sq)
s2_h    = (1 / p) * np.sum((lam * h - lam * m_h) ** 2)
c_JSE   = 1.0 - nu_sq / s2_h
h_JSE   = m_h * np.ones(p) + c_JSE * (h - m_h * np.ones(p))

Sig_JSE = (lambda_sq - ell_sq) * np.outer(h_JSE, h_JSE) \
          / np.dot(h_JSE, h_JSE) \
          + (n / p) * ell_sq * np.eye(p)
inv_JSE = np.linalg.inv(Sig_JSE)
w_JSE   = (inv_JSE @ ones) / (ones.T @ inv_JSE @ ones)
```

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A Complete Portfolio Holdings: Uncorrected ($\Sigma_{\mathbf{PCA}}$)

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
1	PM	+1.1385	23.69	-13.45
2	ED	+1.0787	14.82	-19.63
3	AWK	+1.0748	13.86	-4.34
4	CBOE	+1.0350	18.43	20.27
5	NEM	+1.0061	37.84	92.16
6	DUK	+0.9840	11.57	12.10
7	KR	+0.9834	21.62	-8.45
8	EXE	+0.9718	28.54	1.00
9	CME	+0.9590	15.91	1.25
10	EXC	+0.9209	12.20	7.73
11	LMT	+0.9063	18.61	3.54
12	AEP	+0.8805	13.62	16.69
13	NOC	+0.8711	21.20	49.22
14	ATO	+0.8679	12.43	19.98
15	MO	+0.8654	13.74	24.24
16	CVS	+0.8649	24.12	48.37
17	HSY	+0.8635	27.10	21.42
18	AMD	+0.8359	54.24	205.04
19	CNP	+0.8317	11.06	5.57
20	FE	+0.8294	12.02	19.94
21	DGX	+0.8143	17.13	8.45
22	EQT	+0.8104	31.88	15.36
23	COR	+0.8073	16.41	28.21
24	PEG	+0.8065	14.71	5.00
25	DG	+0.8046	33.61	20.55
26	ETR	+0.7875	13.15	25.87
27	CMS	+0.7858	10.24	6.20
28	PPL	+0.7755	11.58	7.28
29	GNRC	+0.7590	44.86	112.29
30	LHX	+0.7560	16.99	60.43
31	EVRG	+0.7549	12.90	28.20
32	PNW	+0.7532	12.29	-3.16
33	JNJ	+0.7500	14.56	41.80
34	GEV	+0.7408	30.99	91.34
35	NI	+0.7399	14.63	19.69
36	GLW	+0.7374	24.08	138.75
37	AEE	+0.7353	9.76	10.96
38	LNT	+0.7351	11.58	25.08
39	EA	+0.7223	24.02	62.32
40	EBAY	+0.7156	30.36	74.12
41	KVUE	+0.7123	30.97	-80.62
42	FFIV	+0.7037	20.90	17.91
43	COST	+0.7026	13.50	-12.28
44	DTE	+0.7021	9.79	9.65
45	CASY	+0.7015	28.05	34.35
46	D	+0.6876	13.33	29.88
47	KMB	+0.6851	18.27	-19.02
48	GL	+0.6811	21.30	15.40
49	MCK	+0.6794	19.39	26.61
50	RSG	+0.6779	12.59	-20.00
51	PEP	+0.6770	17.11	26.90
52	NFLX	+0.6738	27.17	-1.85
53	KMI	+0.6707	17.04	-6.08
54	MSI	+0.6673	16.24	2.04

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
55	MCD	+0.6658	12.29	-8.31
56	MNST	+0.6637	20.85	32.39
57	CEG	+0.6589	29.11	112.92
58	ACGL	+0.6509	17.07	-10.76
59	ORLY	+0.6502	17.39	13.21
60	CTRA	+0.6421	23.48	-12.79
61	INTU	+0.6388	26.70	18.00
62	EQIX	+0.6350	22.44	0.93
63	FANG	+0.6333	27.56	15.19
64	CAH	+0.6266	18.73	31.36
65	CF	+0.6229	34.10	23.42
66	ABBV	+0.6192	19.84	41.65
67	GILD	+0.6175	22.90	32.99
68	ROL	+0.6170	14.43	2.49
69	ES	+0.6138	17.16	53.20
70	ICE	+0.6092	15.36	-8.99
71	KO	+0.6066	9.96	-7.07
72	O	+0.5967	12.01	12.65
73	AZO	+0.5963	18.93	8.09
74	ABT	+0.5905	16.31	-4.31
75	HII	+0.5887	15.24	55.91
76	ALL	+0.5834	16.75	-0.66
77	CCI	+0.5781	19.90	-1.39
78	MRSH	+0.5747	19.99	-32.70
79	ORCL	+0.5673	56.13	155.99
80	JKHY	+0.5619	16.93	-19.97
81	NDAQ	+0.5613	15.22	30.15
82	HUM	+0.5610	37.18	22.99
83	L	+0.5548	11.69	28.72
84	CB	+0.5528	15.64	-0.39
85	DLTR	+0.5433	27.12	40.81
86	BRK-B	+0.5392	12.64	-18.68
87	CL	+0.5385	15.97	-37.50
88	EOG	+0.5358	24.46	-9.98
89	KDP	+0.5320	31.45	-42.91
90	PODD	+0.5261	36.31	43.51
91	MDLZ	+0.5177	19.54	-14.59
92	ROP	+0.5151	14.58	-33.02
93	CHD	+0.5143	19.00	-30.17
94	EG	+0.5132	15.06	-4.40
95	MSFT	+0.5067	18.26	56.30
96	CNC	+0.5012	69.81	-79.60
97	FISV	+0.5000	29.83	-69.05
98	AIZ	+0.4974	19.95	20.14
99	BK	+0.4942	16.95	63.24
100	CVX	+0.4930	20.02	25.32
101	OXY	+0.4921	28.85	12.72
102	LIN	+0.4918	12.20	-3.36
103	BRO	+0.4827	21.82	-53.50
104	REG	+0.4808	12.94	3.85
105	CTAS	+0.4771	16.53	-19.72
106	PAYX	+0.4759	17.87	-27.53
107	PLTR	+0.4756	49.34	106.86
108	BG	+0.4748	40.14	44.79
109	CAG	+0.4732	21.18	-53.32
110	HIG	+0.4716	15.98	8.99

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
111	MLM	+0.4676	15.64	39.28
112	MSCI	+0.4615	21.10	1.76
113	AMT	+0.4610	19.11	-18.31
114	CTVA	+0.4610	22.75	6.26
115	GIS	+0.4593	15.83	-31.88
116	RMD	+0.4584	19.09	17.50
117	DVN	+0.4552	31.78	13.59
118	AFL	+0.4544	16.52	-2.16
119	CIEN	+0.4537	47.28	209.86
120	LH	+0.4535	21.04	39.57
121	BR	+0.4423	20.53	-5.35
122	PTC	+0.4418	22.19	55.40
123	ADM	+0.4412	23.96	56.97
124	CLX	+0.4390	20.64	-33.72
125	INCY	+0.4373	29.03	87.30
126	PGR	+0.4373	20.04	-40.04
127	BSX	+0.4314	17.95	-4.25
128	PG	+0.4306	14.81	-11.47
129	INVH	+0.4261	15.76	-30.63
130	DLR	+0.4148	18.98	23.25
131	HCA	+0.4131	28.17	62.73
132	FIS	+0.4109	19.35	-31.41
133	ADP	+0.4106	15.38	-8.63
134	PSA	+0.4093	18.14	9.07
135	CPB	+0.4087	20.58	-29.17
136	AIG	+0.4077	18.81	-5.85
137	AMGN	+0.4023	26.21	10.09
138	BDX	+0.3983	33.47	-15.75
139	KHC	+0.3978	17.02	-27.93
140	OKE	+0.3968	19.59	-43.08
141	ALLE	+0.3916	20.54	39.40
142	PCG	+0.3911	28.61	-8.19
143	AON	+0.3869	21.05	-0.12
144	GD	+0.3858	16.05	50.03
145	OTIS	+0.3839	20.70	-2.39
146	AME	+0.3809	14.29	20.58
147	META	+0.3781	27.46	59.92
148	HWM	+0.3778	27.07	76.04
149	NEE	+0.3774	28.49	52.16
150	CSCO	+0.3685	22.67	44.78
151	CRWD	+0.3646	34.87	44.98
152	AXON	+0.3592	40.25	43.48
153	MKC	+0.3587	18.98	-21.65
154	ELV	+0.3587	38.93	-35.84
155	CDNS	+0.3541	31.41	35.78
156	DPZ	+0.3510	19.21	-33.10
157	ECL	+0.3505	17.47	28.67
158	COP	+0.3503	30.99	-4.40
159	BMY	+0.3502	24.89	-13.53
160	ERIE	+0.3465	22.62	-21.70
161	NWSA	+0.3462	22.18	-7.16
162	GE	+0.3430	24.30	85.12
163	EFX	+0.3405	21.89	-19.56
164	AJG	+0.3349	22.16	-28.48
165	CINF	+0.3340	18.75	31.59
166	CMCSA	+0.3316	16.91	-28.18

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
167	CPT	+0.3311	18.16	-15.13
168	NWS	+0.3266	22.16	-8.23
169	AVB	+0.3248	18.46	-15.54
170	FDS	+0.3240	33.91	-75.80
171	RTX	+0.3160	23.92	71.93
172	ADSK	+0.3120	26.89	29.00
173	PSKY	+0.3049	69.21	90.73
174	CSGP	+0.2999	30.50	-11.88
175	DASH	+0.2955	31.29	64.62
176	DRI	+0.2940	29.12	-11.77
177	MAA	+0.2920	19.28	-32.73
178	APH	+0.2910	20.83	113.25
179	AMCR	+0.2897	22.83	-30.49
180	MRK	+0.2887	31.25	15.64
181	EQR	+0.2810	17.98	-13.99
182	HRL	+0.2803	23.58	-40.87
183	CSX	+0.2788	21.74	52.08
184	CHRW	+0.2742	30.61	70.95
185	GOOG	+0.2702	28.38	93.66
186	GOOGL	+0.2638	29.12	95.76
187	NSC	+0.2634	20.42	47.20
188	MDT	+0.2593	19.31	22.21
189	COHR	+0.2587	46.76	148.08
190	JPM	+0.2577	17.60	41.34
191	ETN	+0.2572	21.87	52.54
192	CI	+0.2549	28.61	-19.27
193	GDDY	+0.2503	30.64	-66.40
194	FAST	+0.2491	24.90	12.89
195	MCO	+0.2478	20.41	21.07
196	INTC	+0.2474	57.38	141.78
197	EW	+0.2432	21.92	-1.76
198	LYV	+0.2386	28.66	28.03
199	EIX	+0.2335	28.55	5.51
200	LDOS	+0.2334	24.06	52.81
201	NVR	+0.2326	22.05	8.10
202	GEN	+0.2270	28.71	16.39
203	CBRE	+0.2263	22.82	58.66
204	BALL	+0.2262	20.85	-3.50
205	J	+0.2255	18.68	55.09
206	DOC	+0.2234	20.61	14.89
207	JBL	+0.2207	35.09	76.02
208	IBM	+0.2206	28.37	58.69
209	AMAT	+0.2175	36.88	86.65
210	LLY	+0.2158	48.73	-5.18
211	HUBB	+0.2157	21.12	36.94
212	AVGO	+0.2134	30.63	124.62
213	BLK	+0.2105	18.89	44.64
214	FRT	+0.2049	18.65	16.19
215	HD	+0.2035	21.39	16.03
216	APA	+0.2022	38.71	85.09
217	DELL	+0.2012	37.45	108.28
218	RJF	+0.1988	20.50	32.68
219	AMP	+0.1974	20.54	4.72
220	DIS	+0.1906	27.37	42.85
221	HON	+0.1883	21.91	16.52
222	IRM	+0.1865	22.90	38.03

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
223	HSIC	+0.1863	20.72	-6.26
224	ITW	+0.1823	17.14	4.78
225	BA	+0.1800	24.85	42.61
226	EXR	+0.1707	23.49	15.54
227	MA	+0.1687	20.59	12.91
228	DVA	+0.1681	20.63	-17.56
229	MPC	+0.1646	27.53	73.41
230	BKNG	+0.1607	23.36	11.43
231	BKR	+0.1596	31.56	55.20
232	RL	+0.1574	24.64	85.19
233	EL	+0.1573	34.54	112.84
234	HAL	+0.1560	48.34	58.70
235	EME	+0.1516	25.31	120.51
236	NTAP	+0.1512	25.43	56.39
237	ROK	+0.1494	28.50	73.96
238	DE	+0.1490	21.22	5.04
239	KIM	+0.1489	19.38	20.91
240	MOS	+0.1476	32.86	9.55
241	MET	+0.1461	20.25	9.84
242	FCX	+0.1419	42.72	27.33
243	CAT	+0.1331	26.06	108.64
244	REGN	+0.1327	42.50	-2.69
245	ESS	+0.1320	23.60	-5.71
246	AOS	+0.1302	21.68	11.76
247	PRU	+0.1288	19.29	1.24
248	PFE	+0.1269	34.64	24.29
249	OMC	+0.1268	23.56	12.86
250	COO	+0.1263	32.18	-18.00
251	JCI	+0.1219	21.43	65.41
252	GRMN	+0.1167	24.64	46.61
253	HPE	+0.1154	30.94	77.27
254	APD	+0.1143	22.23	-8.54
255	AMZN	+0.1086	22.55	32.53
256	AKAM	+0.1073	24.45	-13.49
257	LOW	+0.1035	26.44	20.16
258	HAS	+0.0988	26.28	49.01
259	PWR	+0.0923	25.79	85.40
260	LEN	+0.0917	30.62	37.18
261	CVNA	+0.0886	45.25	85.19
262	ARES	+0.0879	29.91	-3.11
263	ANET	+0.0841	52.83	146.84
264	PFG	+0.0795	20.75	17.95
265	FOXA	+0.0788	26.41	37.18
266	HLT	+0.0778	23.16	38.36
267	IDXX	+0.0768	39.39	79.74
268	MPWR	+0.0765	36.22	126.41
269	SATS	+0.0726	169.10	334.00
270	DHI	+0.0722	33.65	50.38
271	NOW	+0.0707	26.19	-4.02
272	C	+0.0661	25.34	75.77
273	IBKR	+0.0625	31.16	98.93
274	BIIB	+0.0623	36.76	48.90
275	NVDA	+0.0611	30.32	104.62
276	PNR	+0.0597	21.00	38.14
277	PHM	+0.0591	31.58	35.31
278	IT	+0.0589	49.15	-90.75

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
279	GS	+0.0574	23.25	73.69
280	BEN	+0.0548	23.29	43.86
281	NUE	+0.0546	30.65	38.37
282	EXPD	+0.0519	24.84	16.51
283	CRM	+0.0517	25.31	-10.36
284	PLD	+0.0499	24.94	45.11
285	FTNT	+0.0486	43.98	-28.26
286	CTSH	+0.0477	24.59	-11.90
287	ADBE	+0.0460	25.81	-8.87
288	FOX	+0.0459	26.76	29.18
289	AAPL	+0.0452	30.91	46.63
290	PKG	+0.0435	26.59	22.25
291	GM	+0.0402	35.52	81.66
292	MS	+0.0399	22.01	70.64
293	PCAR	+0.0384	24.75	17.27
294	ROST	+0.0382	28.38	23.97
295	CMG	+0.0268	34.27	-43.83
296	NDSN	+0.0249	23.76	45.31
297	NTRS	+0.0245	25.88	65.89
298	ARE	+0.0236	28.61	10.68
299	FIX	+0.0194	43.37	188.52
300	CMI	+0.0174	24.33	73.83
301	EMR	+0.0167	24.62	46.90
302	EXPE	+0.0133	27.56	63.21
303	CRH	+0.0131	30.40	53.67
304	CPRT	+0.0112	27.34	-61.95
305	IVZ	+0.0095	30.90	106.36
306	BF-B	+0.0073	37.47	-32.70
307	QCOM	+0.0064	28.59	27.96
308	PH	+0.0059	21.29	50.33
309	ACN	+0.0054	29.47	-32.62
310	BAC	+0.0053	23.08	57.19
311	FTV	-0.0035	24.51	-10.48
312	GWW	-0.0046	24.95	-9.36
313	DOV	-0.0120	21.92	8.97
314	HST	-0.0146	25.67	38.47
315	COF	-0.0152	25.47	41.92
316	AVY	-0.0175	27.30	13.77
317	MTB	-0.0214	23.26	21.70
318	GPC	-0.0222	26.27	25.37
319	KEYS	-0.0244	24.06	31.27
320	DAL	-0.0346	36.02	81.04
321	PSX	-0.0511	35.05	58.17
322	PANW	-0.0515	32.19	39.91
323	PYPL	-0.0562	30.36	13.65
324	POOL	-0.0569	28.05	2.63
325	MMM	-0.0574	27.07	42.29
326	DXCM	-0.0617	40.40	0.15
327	CPAY	-0.0640	25.67	-25.81
328	ADI	-0.0651	30.07	42.33
329	LVS	-0.0682	38.52	100.58
330	BX	-0.0706	31.21	33.62
331	A	-0.0717	31.90	68.19
332	FITB	-0.0796	24.81	40.36
333	BXP	-0.0811	32.40	29.61
334	IFF	-0.0820	29.33	-26.58

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
335	IQV	-0.0838	47.10	83.21
336	KLAC	-0.0843	37.55	111.12
337	APO	-0.0854	32.53	-10.48
338	MAS	-0.0863	27.72	25.56
339	DDOG	-0.0871	40.88	91.17
340	CDW	-0.0877	28.46	1.48
341	AXP	-0.0924	28.36	61.12
342	APP	-0.0939	60.44	177.68
343	DHR	-0.0991	37.04	27.73
344	NRG	-0.0996	54.77	101.54
345	ISRG	-0.1052	39.98	15.17
346	MTD	-0.1085	28.16	60.11
347	LII	-0.1115	32.28	-9.84
348	ON	-0.1138	51.67	58.38
349	DD	-0.1170	26.55	43.60
350	LRCX	-0.1205	39.55	157.05
351	MAR	-0.1210	24.54	27.58
352	PNC	-0.1211	25.70	35.20
353	LUV	-0.1251	39.53	44.95
354	HPQ	-0.1332	33.68	23.72
355	MU	-0.1333	55.12	216.28
356	HOOD	-0.1333	57.95	223.67
357	NCLH	-0.1406	37.22	64.85
358	CHTR	-0.1409	41.21	-79.55
359	LITE	-0.1606	46.41	229.60
360	RF	-0.1637	26.95	42.21
361	KKR	-0.1740	32.89	14.73
362	MGM	-0.1742	34.12	8.65
363	F	-0.1758	34.54	71.74
364	IEX	-0.1898	27.88	-3.75
365	CRL	-0.1916	51.05	117.56
366	JBHT	-0.1980	37.87	52.18
367	ABNB	-0.2023	32.22	9.67
368	GEHC	-0.2072	27.28	26.34
369	PPG	-0.2074	27.92	4.04
370	HBAN	-0.2096	27.35	25.26
371	CFG	-0.2303	27.15	72.33
372	KEY	-0.2350	28.23	42.85
373	BBY	-0.2356	39.04	50.07
374	FDX	-0.2486	30.71	29.33
375	MCHP	-0.2502	44.24	68.19
376	BLDR	-0.2503	44.27	9.99
377	AES	-0.2788	45.09	87.10
378	MRNA	-0.2890	57.59	9.04
379	APTV	-0.3075	31.41	87.82
380	GPN	-0.3096	32.15	39.66
381	ALB	-0.3480	52.39	134.41
382	LYB	-0.3658	42.58	-29.71
383	RVTY	-0.3685	41.02	14.05
384	IR	-0.3807	32.42	13.78
385	CCL	-0.4083	41.09	96.20
386	NKE	-0.4116	39.59	41.99
387	LULU	-0.4175	51.04	-71.74
388	NXPI	-0.4181	36.86	29.24
389	EPAM	-0.4349	41.44	4.61
390	DOW	-0.4399	48.23	-22.70

#	**Stock**	**Weight (%)**	**Ann. σ (%)**	**Ann. $E[R]$ (%)**
391	ODFL	-0.4460	34.63	-12.40
392	CARR	-0.4918	42.23	-1.71
393	IP	-0.5094	41.17	14.51
394	RCL	-0.5429	43.08	86.82
395	BAX	-0.5442	45.04	-44.66
396	FSLR	-0.6193	64.91	123.51
397	DECK	-0.6392	53.75	-34.56
398	FICO	-0.7217	58.58	-17.88
399	ALGN	-0.9069	56.10	-45.89
400	COIN	-1.0143	81.56	132.48

Table 1: Complete uncorrected (Σ_{PCA}) minimum variance portfolio holdings, all 400 stocks, sorted by weight descending.

B Complete Portfolio Holdings: JSE-Corrected ($\Sigma_{\mathbf{JSE}}$)

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
1	PM	+1.6351	23.69	-13.45
2	ED	+1.5419	14.82	-19.63
3	AWK	+1.5358	13.86	-4.34
4	CBOE	+1.4738	18.43	20.27
5	NEM	+1.4287	37.84	92.16
6	DUK	+1.3942	11.57	12.10
7	KR	+1.3933	21.62	-8.45
8	EXE	+1.3752	28.54	1.00
9	CME	+1.3553	15.91	1.25
10	EXC	+1.2959	12.20	7.73
11	LMT	+1.2731	18.61	3.54
12	AEP	+1.2329	13.62	16.69
13	NOC	+1.2183	21.20	49.22
14	ATO	+1.2133	12.43	19.98
15	MO	+1.2094	13.74	24.24
16	CVS	+1.2086	24.12	48.37
17	HSY	+1.2064	27.10	21.42
18	AMD	+1.1633	54.24	205.04
19	CNP	+1.1568	11.06	5.57
20	FE	+1.1532	12.02	19.94
21	DGX	+1.1296	17.13	8.45
22	EQT	+1.1236	31.88	15.36
23	COR	+1.1188	16.41	28.21
24	PEG	+1.1175	14.71	5.00
25	DG	+1.1146	33.61	20.55
26	ETR	+1.0880	13.15	25.87
27	CMS	+1.0853	10.24	6.20
28	PPL	+1.0692	11.58	7.28
29	GNRC	+1.0435	44.86	112.29
30	LHX	+1.0388	16.99	60.43
31	EVRG	+1.0371	12.90	28.20
32	PNW	+1.0345	12.29	-3.16
33	JNJ	+1.0295	14.56	41.80
34	GEV	+1.0151	30.99	91.34
35	NI	+1.0137	14.63	19.69
36	GLW	+1.0098	24.08	138.75
37	AEE	+1.0066	9.76	10.96
38	LNT	+1.0062	11.58	25.08
39	EA	+0.9863	24.02	62.32
40	EBAY	+0.9759	30.36	74.12
41	KVUE	+0.9707	30.97	-80.62
42	FFIV	+0.9573	20.90	17.91
43	COST	+0.9556	13.50	-12.28
44	DTE	+0.9547	9.79	9.65
45	CASY	+0.9538	28.05	34.35
46	D	+0.9322	13.33	29.88
47	KMB	+0.9284	18.27	-19.02
48	GL	+0.9221	21.30	15.40
49	MCK	+0.9194	19.39	26.61
50	RSG	+0.9171	12.59	-20.00
51	PEP	+0.9157	17.11	26.90
52	NFLX	+0.9107	27.17	-1.85
53	KMI	+0.9058	17.04	-6.08
54	MSI	+0.9006	16.24	2.04

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
55	MCD	+0.8982	12.29	-8.31
56	MNST	+0.8949	20.85	32.39
57	CEG	+0.8874	29.11	112.92
58	ACGL	+0.8750	17.07	-10.76
59	ORLY	+0.8739	17.39	13.21
60	CTRA	+0.8613	23.48	-12.79
61	INTU	+0.8562	26.70	18.00
62	EQIX	+0.8502	22.44	0.93
63	FANG	+0.8476	27.56	15.19
64	CAH	+0.8370	18.73	31.36
65	CF	+0.8313	34.10	23.42
66	ABBV	+0.8256	19.84	41.65
67	GILD	+0.8229	22.90	32.99
68	ROL	+0.8222	14.43	2.49
69	ES	+0.8172	17.16	53.20
70	ICE	+0.8099	15.36	-8.99
71	KO	+0.8058	9.96	-7.07
72	O	+0.7906	12.01	12.65
73	AZO	+0.7898	18.93	8.09
74	ABT	+0.7808	16.31	-4.31
75	HII	+0.7780	15.24	55.91
76	ALL	+0.7698	16.75	-0.66
77	CCI	+0.7615	19.90	-1.39
78	MRSH	+0.7562	19.99	-32.70
79	ORCL	+0.7447	56.13	155.99
80	JKHY	+0.7362	16.93	-19.97
81	NDAQ	+0.7352	15.22	30.15
82	HUM	+0.7349	37.18	22.99
83	L	+0.7251	11.69	28.72
84	CB	+0.7221	15.64	-0.39
85	DLTR	+0.7073	27.12	40.81
86	BRK-B	+0.7009	12.64	-18.68
87	CL	+0.6998	15.97	-37.50
88	EOG	+0.6955	24.46	-9.98
89	KDP	+0.6896	31.45	-42.91
90	PODD	+0.6804	36.31	43.51
91	MDLZ	+0.6674	19.54	-14.59
92	ROP	+0.6632	14.58	-33.02
93	CHD	+0.6620	19.00	-30.17
94	EG	+0.6603	15.06	-4.40
95	MSFT	+0.6502	18.26	56.30
96	CNC	+0.6417	69.81	-79.60
97	FISV	+0.6397	29.83	-69.05
98	AIZ	+0.6356	19.95	20.14
99	BK	+0.6307	16.95	63.24
100	CVX	+0.6288	20.02	25.32
101	OXY	+0.6275	28.85	12.72
102	LIN	+0.6270	12.20	-3.36
103	BRO	+0.6127	21.82	-53.50
104	REG	+0.6097	12.94	3.85
105	CTAS	+0.6041	16.53	-19.72
106	PAYX	+0.6021	17.87	-27.53
107	PLTR	+0.6017	49.34	106.86
108	BG	+0.6005	40.14	44.79
109	CAG	+0.5980	21.18	-53.32
110	HIG	+0.5955	15.98	8.99

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
111	MLM	+0.5892	15.64	39.28
112	MSCI	+0.5798	21.10	1.76
113	AMT	+0.5790	19.11	-18.31
114	CTVA	+0.5789	22.75	6.26
115	GIS	+0.5763	15.83	-31.88
116	RMD	+0.5748	19.09	17.50
117	DVN	+0.5699	31.78	13.59
118	AFL	+0.5687	16.52	-2.16
119	CIEN	+0.5675	47.28	209.86
120	LH	+0.5672	21.04	39.57
121	BR	+0.5497	20.53	-5.35
122	PTC	+0.5489	22.19	55.40
123	ADM	+0.5480	23.96	56.97
124	CLX	+0.5447	20.64	-33.72
125	INCY	+0.5420	29.03	87.30
126	PGR	+0.5420	20.04	-40.04
127	BSX	+0.5329	17.95	-4.25
128	PG	+0.5315	14.81	-11.47
129	INVH	+0.5246	15.76	-30.63
130	DLR	+0.5068	18.98	23.25
131	HCA	+0.5043	28.17	62.73
132	FIS	+0.5008	19.35	-31.41
133	ADP	+0.5004	15.38	-8.63
134	PSA	+0.4983	18.14	9.07
135	CPB	+0.4973	20.58	-29.17
136	AIG	+0.4958	18.81	-5.85
137	AMGN	+0.4875	26.21	10.09
138	BDX	+0.4812	33.47	-15.75
139	KHC	+0.4804	17.02	-27.93
140	OKE	+0.4789	19.59	-43.08
141	ALLE	+0.4707	20.54	39.40
142	PCG	+0.4700	28.61	-8.19
143	AON	+0.4634	21.05	-0.12
144	GD	+0.4618	16.05	50.03
145	OTIS	+0.4588	20.70	-2.39
146	AME	+0.4540	14.29	20.58
147	META	+0.4497	27.46	59.92
148	HWM	+0.4492	27.07	76.04
149	NEE	+0.4485	28.49	52.16
150	CSCO	+0.4348	22.67	44.78
151	CRWD	+0.4286	34.87	44.98
152	AXON	+0.4202	40.25	43.48
153	MKC	+0.4195	18.98	-21.65
154	ELV	+0.4195	38.93	-35.84
155	CDNS	+0.4123	31.41	35.78
156	DPZ	+0.4075	19.21	-33.10
157	ECL	+0.4066	17.47	28.67
158	COP	+0.4064	30.99	-4.40
159	BMY	+0.4062	24.89	-13.53
160	ERIE	+0.4004	22.62	-21.70
161	NWSA	+0.4000	22.18	-7.16
162	GE	+0.3951	24.30	85.12
163	EFX	+0.3911	21.89	-19.56
164	AJG	+0.3824	22.16	-28.48
165	CINF	+0.3809	18.75	31.59
166	CMCSA	+0.3773	16.91	-28.18

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
167	CPT	+0.3765	18.16	-15.13
168	NWS	+0.3695	22.16	-8.23
169	AVB	+0.3667	18.46	-15.54
170	FDS	+0.3653	33.91	-75.80
171	RTX	+0.3528	23.92	71.93
172	ADSK	+0.3467	26.89	29.00
173	PSKY	+0.3356	69.21	90.73
174	CSGP	+0.3278	30.50	-11.88
175	DASH	+0.3210	31.29	64.62
176	DRI	+0.3186	29.12	-11.77
177	MAA	+0.3155	19.28	-32.73
178	APH	+0.3139	20.83	113.25
179	AMCR	+0.3118	22.83	-30.49
180	MRK	+0.3103	31.25	15.64
181	EQR	+0.2984	17.98	-13.99
182	HRL	+0.2972	23.58	-40.87
183	CSX	+0.2950	21.74	52.08
184	CHRW	+0.2877	30.61	70.95
185	GOOG	+0.2815	28.38	93.66
186	GOOGL	+0.2715	29.12	95.76
187	NSC	+0.2709	20.42	47.20
188	MDT	+0.2645	19.31	22.21
189	COHR	+0.2636	46.76	148.08
190	JPM	+0.2619	17.60	41.34
191	ETN	+0.2613	21.87	52.54
192	CI	+0.2577	28.61	-19.27
193	GDDY	+0.2505	30.64	-66.40
194	FAST	+0.2487	24.90	12.89
195	MCO	+0.2466	20.41	21.07
196	INTC	+0.2459	57.38	141.78
197	EW	+0.2395	21.92	-1.76
198	LYV	+0.2323	28.66	28.03
199	EIX	+0.2243	28.55	5.51
200	LDOS	+0.2241	24.06	52.81
201	NVR	+0.2229	22.05	8.10
202	GEN	+0.2141	28.71	16.39
203	CBRE	+0.2131	22.82	58.66
204	BALL	+0.2129	20.85	-3.50
205	J	+0.2119	18.68	55.09
206	DOC	+0.2085	20.61	14.89
207	JBL	+0.2044	35.09	76.02
208	IBM	+0.2041	28.37	58.69
209	AMAT	+0.1994	36.88	86.65
210	LLY	+0.1968	48.73	-5.18
211	HUBB	+0.1965	21.12	36.94
212	AVGO	+0.1929	30.63	124.62
213	BLK	+0.1884	18.89	44.64
214	FRT	+0.1797	18.65	16.19
215	HD	+0.1775	21.39	16.03
216	APA	+0.1755	38.71	85.09
217	DELL	+0.1740	37.45	108.28
218	RJF	+0.1701	20.50	32.68
219	AMP	+0.1680	20.54	4.72
220	DIS	+0.1575	27.37	42.85
221	HON	+0.1539	21.91	16.52
222	IRM	+0.1510	22.90	38.03

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
223	HSIC	+0.1506	20.72	-6.26
224	ITW	+0.1445	17.14	4.78
225	BA	+0.1408	24.85	42.61
226	EXR	+0.1264	23.49	15.54
227	MA	+0.1232	20.59	12.91
228	DVA	+0.1222	20.63	-17.56
229	MPC	+0.1169	27.53	73.41
230	BKNG	+0.1108	23.36	11.43
231	BKR	+0.1091	31.56	55.20
232	RL	+0.1057	24.64	85.19
233	EL	+0.1054	34.54	112.84
234	HAL	+0.1035	48.34	58.70
235	EME	+0.0966	25.31	120.51
236	NTAP	+0.0960	25.43	56.39
237	ROK	+0.0932	28.50	73.96
238	DE	+0.0926	21.22	5.04
239	KIM	+0.0924	19.38	20.91
240	MOS	+0.0903	32.86	9.55
241	MET	+0.0881	20.25	9.84
242	FCX	+0.0815	42.72	27.33
243	CAT	+0.0678	26.06	108.64
244	REGN	+0.0671	42.50	-2.69
245	ESS	+0.0660	23.60	-5.71
246	AOS	+0.0632	21.68	11.76
247	PRU	+0.0611	19.29	1.24
248	PFE	+0.0580	34.64	24.29
249	OMC	+0.0580	23.56	12.86
250	COO	+0.0571	32.18	-18.00
251	JCI	+0.0502	21.43	65.41
252	GRMN	+0.0422	24.64	46.61
253	HPE	+0.0402	30.94	77.27
254	APD	+0.0385	22.23	-8.54
255	AMZN	+0.0295	22.55	32.53
256	AKAM	+0.0276	24.45	-13.49
257	LOW	+0.0216	26.44	20.16
258	HAS	+0.0142	26.28	49.01
259	PWR	+0.0042	25.79	85.40
260	LEN	+0.0033	30.62	37.18
261	CVNA	-0.0016	45.25	85.19
262	ARES	-0.0027	29.91	-3.11
263	ANET	-0.0086	52.83	146.84
264	PFG	-0.0158	20.75	17.95
265	FOXA	-0.0168	26.41	37.18
266	HLT	-0.0184	23.16	38.36
267	IDXX	-0.0200	39.39	79.74
268	MPWR	-0.0205	36.22	126.41
269	SATS	-0.0266	169.10	334.00
270	DHI	-0.0271	33.65	50.38
271	NOW	-0.0295	26.19	-4.02
272	C	-0.0367	25.34	75.77
273	IBKR	-0.0423	31.16	98.93
274	BIIB	-0.0427	36.76	48.90
275	NVDA	-0.0444	30.32	104.62
276	PNR	-0.0466	21.00	38.14
277	PHM	-0.0476	31.58	35.31
278	IT	-0.0480	49.15	-90.75

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
279	GS	-0.0502	23.25	73.69
280	BEN	-0.0542	23.29	43.86
281	NUE	-0.0546	30.65	38.37
282	EXPD	-0.0588	24.84	16.51
283	CRM	-0.0591	25.31	-10.36
284	PLD	-0.0620	24.94	45.11
285	FTNT	-0.0640	43.98	-28.26
286	CTSH	-0.0654	24.59	-11.90
287	ADBE	-0.0680	25.81	-8.87
288	FOX	-0.0681	26.76	29.18
289	AAPL	-0.0693	30.91	46.63
290	PKG	-0.0720	26.59	22.25
291	GM	-0.0770	35.52	81.66
292	MS	-0.0775	22.01	70.64
293	PCAR	-0.0799	24.75	17.27
294	ROST	-0.0802	28.38	23.97
295	CMG	-0.0980	34.27	-43.83
296	NDSN	-0.1009	23.76	45.31
297	NTRS	-0.1016	25.88	65.89
298	ARE	-0.1030	28.61	10.68
299	FIX	-0.1096	43.37	188.52
300	CMI	-0.1127	24.33	73.83
301	EMR	-0.1136	24.62	46.90
302	EXPE	-0.1190	27.56	63.21
303	CRH	-0.1193	30.40	53.67
304	CPRT	-0.1223	27.34	-61.95
305	IVZ	-0.1250	30.90	106.36
306	BF-B	-0.1284	37.47	-32.70
307	QCOM	-0.1297	28.59	27.96
308	PH	-0.1306	21.29	50.33
309	ACN	-0.1314	29.47	-32.62
310	BAC	-0.1315	23.08	57.19
311	FTV	-0.1452	24.51	-10.48
312	GWW	-0.1470	24.95	-9.36
313	DOV	-0.1585	21.92	8.97
314	HST	-0.1626	25.67	38.47
315	COF	-0.1635	25.47	41.92
316	AVY	-0.1671	27.30	13.77
317	MTB	-0.1732	23.26	21.70
318	GPC	-0.1744	26.27	25.37
319	KEYS	-0.1778	24.06	31.27
320	DAL	-0.1936	36.02	81.04
321	PSX	-0.2193	35.05	58.17
322	PANW	-0.2200	32.19	39.91
323	PYPL	-0.2273	30.36	13.65
324	POOL	-0.2285	28.05	2.63
325	MMM	-0.2292	27.07	42.29
326	DXCM	-0.2359	40.40	0.15
327	CPAY	-0.2396	25.67	-25.81
328	ADI	-0.2412	30.07	42.33
329	LVS	-0.2461	38.52	100.58
330	BX	-0.2497	31.21	33.62
331	A	-0.2514	31.90	68.19
332	FITB	-0.2638	24.81	40.36
333	BXP	-0.2661	32.40	29.61
334	IFF	-0.2675	29.33	-26.58

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
335	IQV	-0.2703	47.10	83.21
336	KLAC	-0.2711	37.55	111.12
337	APO	-0.2729	32.53	-10.48
338	MAS	-0.2742	27.72	25.56
339	DDOG	-0.2755	40.88	91.17
340	CDW	-0.2765	28.46	1.48
341	AXP	-0.2838	28.36	61.12
342	APP	-0.2861	60.44	177.68
343	DHR	-0.2943	37.04	27.73
344	NRG	-0.2950	54.77	101.54
345	ISRG	-0.3037	39.98	15.17
346	MTD	-0.3089	28.16	60.11
347	LII	-0.3136	32.28	-9.84
348	ON	-0.3171	51.67	58.38
349	DD	-0.3221	26.55	43.60
350	LRCX	-0.3277	39.55	157.05
351	MAR	-0.3284	24.54	27.58
352	PNC	-0.3285	25.70	35.20
353	LUV	-0.3348	39.53	44.95
354	HPQ	-0.3474	33.68	23.72
355	MU	-0.3476	55.12	216.28
356	HOOD	-0.3476	57.95	223.67
357	NCLH	-0.3589	37.22	64.85
358	CHTR	-0.3594	41.21	-79.55
359	LITE	-0.3901	46.41	229.60
360	RF	-0.3949	26.95	42.21
361	KKR	-0.4109	32.89	14.73
362	MGM	-0.4112	34.12	8.65
363	F	-0.4138	34.54	71.74
364	IEX	-0.4356	27.88	-3.75
365	CRL	-0.4384	51.05	117.56
366	JBHT	-0.4485	37.87	52.18
367	ABNB	-0.4551	32.22	9.67
368	GEHC	-0.4628	27.28	26.34
369	PPG	-0.4630	27.92	4.04
370	HBAN	-0.4665	27.35	25.26
371	CFG	-0.4987	27.15	72.33
372	KEY	-0.5061	28.23	42.85
373	BBY	-0.5070	39.04	50.07
374	FDX	-0.5273	30.71	29.33
375	MCHP	-0.5298	44.24	68.19
376	BLDR	-0.5300	44.27	9.99
377	AES	-0.5744	45.09	87.10
378	MRNA	-0.5903	57.59	9.04
379	APTV	-0.6190	31.41	87.82
380	GPN	-0.6224	32.15	39.66
381	ALB	-0.6823	52.39	134.41
382	LYB	-0.7100	42.58	-29.71
383	RVTY	-0.7143	41.02	14.05
384	IR	-0.7332	32.42	13.78
385	CCL	-0.7762	41.09	96.20
386	NKE	-0.7814	39.59	41.99
387	LULU	-0.7905	51.04	-71.74
388	NXPI	-0.7915	36.86	29.24
389	EPAM	-0.8176	41.44	4.61
390	DOW	-0.8254	48.23	-22.70

#	Stock	Weight (%)	Ann. σ (%)	Ann. $E[R]$ (%)
391	ODFL	-0.8350	34.63	-12.40
392	CARR	-0.9064	42.23	-1.71
393	IP	-0.9339	41.17	14.51
394	RCL	-0.9861	43.08	86.82
395	BAX	-0.9881	45.04	-44.66
396	FSLR	-1.1052	64.91	123.51
397	DECK	-1.1363	53.75	-34.56
398	FICO	-1.2648	58.58	-17.88
399	ALGN	-1.5536	56.10	-45.89
400	COIN	-1.7210	81.56	132.48

Table 2: Complete JSE-corrected (Σ_{JSE}) minimum variance portfolio holdings, all 400 stocks, sorted by weight descending.